David J. Setton

Brinson Prize Fellow

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Summary

Research: Observational galaxy formation and evolution; the quenching of massive galaxies; the co-evolution of galaxies and supermassive black holes; spectral energy distribution modeling with UV-to-IR spectrophotometry; galaxy structure modeling; optical, NIR, and FIR spectroscopy

Grants: Over **730k** in grants and awards secured over the past 5 years

Presentations: **20** invited talks, **9** submitted talks

Publications: **8** first author, **8** second author, **63** total

Students supervised: **6** graduate students, **5** undergraduates, principal advisor of **4** student-led papers

Collaborations: UNCOVER, RUBIES, SQuIGG$\vec{L}$E, MINERVA, PFS-SSP

Observing: PI/Co-I of several JWST, ALMA, HST, VLA, Chandra, Magellan, Keck, and Gemini programs

Professional Appointments

Brinson Prize Fellow at Princeton University — Sep 2023 – Present
Faculty Mentor: Jenny Greene

Education

University of Pittsburgh, MS, PhD in Physics — Aug 2017 – July 2023
Thesis Advisor: Rachel Bezanson

University of Arizona, BS in Physics, BS in Astronomy — Aug 2013 – May 2017
Thesis Advisor: Gurtina Besla

Observing Programs

PI, Co-PI, and Admin-PI Programs:

*James Webb Space Telescope (**3** programs, 38.4 hours):* Cycle 4 GO #8607, Cycle 4 GO #8915, Cycle 3 GO #6719

*Atacama Large Millimeter/sub-millimeter Array (**5** programs, 107.5 hours):* Cycle 11: 2024.1.01064.S; Cycle 10: 2023.1.01012.S; Cycle 9: 2022.1.00604.S; Cycle 8: 2021.1.01535.S, 2021.1.00988.S

*Hubble Space Telescope (**1** program, 409 SNAP Orbits)*: Cycle 30 #17110

Magellan Telescopes: **5** nights using FIRE

Co-I Program Summary:

James Webb Space Telescope: **8** programs, **422.7** hours; *Atacama Large Millimeter/submillimeter Array*: **11** programs, **182.7** hours; *Hubble Space Telescope*: **2** programs, **97** orbits; *NOEMA*: **1** program, **5** hours; *CHANDRA*: **1** program, **48** hours; *Very Large Array*: **2** programs, **66** hours

Grants and Awards

Grants: Brinson Prize Fellowship (**$330,000**); JWST GO #8607 (**Budget Pending**); JWST GO #8915 (**Budget Pending**); JWST GO #6719 (**$128,878**); HST GO #17110 (**$202,893**); ALMA Student Observing Support (∼**$35k**); University of Pittsburgh Graduate Fellowships (∼**$38k**)

Awards: Myron P. Garfunkel Excellence in Graduate Student Teaching Award; Martin and Beate Block Winter Award for Promising Young Physicists; University of Pittsburgh Department of Astronomy and Physics “3 Minute Thesis” Department Competition Winner

Students Supervised

Graduate Students:

Yunchong Zhang (University of Pittsburgh, **2** supervised papers, **1** supervised **A-rated ALMA program**)
Jared Siegel (Princeton University, **1** supervised second-author paper, **co-PI of JWST Cycle 4 program**)
Yilun Ma (Princeton University, **3** supervised papers, multiple successful telescope proposals)
Abby Mintz (Princeton University, **1** supervised second-author paper)
Helena Treiber (Princeton University)
Kaitlyn Shavelle (Princeton University)

Undergraduate Students:

Maggie Verrico (University of Pittsburgh, **1** supervised second-author paper, now graduate student at UIUC)
Anika Kumar (University of Pittsburgh, **1** supervised second-author research note, now graduate student at RIT)
Erin Stumbaugh (University of Pittsburgh)
Belinda Wu (Princeton University, supervised junior thesis, now graduate student at the National Taiwan University)
Hy Troung (Princeton University, supervised summer research project, now graduate student at SDSU)
Austin Guo (University of Pittsburgh, currently supervising senior thesis)

Selected Professional Talks

* = invited	
*Joining efforts to solve JWST mysteries in the distant Universe (**LRD review**)	*Sesto, IT*; Jan 2026
*AAS 247: PRIMA Special Session	*Phoenix, AZ*; Jan 2026
*University of British Columbia Astronomy Colloquium	*Vancouver, BC*; Oct 2025
*University of Kentucky Astronomy Seminar	*Lexington, KY*; Sep 2025
Massive Black Holes across Cosmic Time	*Cambridge, UK*; Sep 2025
Galaxy memoirs: inferring their past from their present	*Buzios, BR*; Aug 2025
New Data that Challenge Underlying Assumptions in Galaxy Evolution	*Bar Harbor, Maine*; July 2025
*PRIMA and the Future of Far-Infrared Astronomy	*Pasadena, CA*; May 2025
Big Galaxies, Big Problems Lorentz Center Meeting (**organizer**)	*Leiden, NL*; Apr 2025
*University of Washington ALMA Workshop	*Seattle, WA*; Mar 2025
*University of Pennsylvania AstroLunch	*Philadelphia, PA*; Mar 2025
*York University Seminar	*Toronto, ON*; Nov 2024
*University of Toronto TASTY Seminar	*Toronto, ON*; Nov 2024
Galaxies and Black Holes in the Early Universe	*New Haven, CT*; Oct 2024
*JHU/STScI Galaxy+AGN Journal Club	*Baltimore, MD*; Apr 2024
*Yale Galaxy Lunch	*New Haven, CT*; Jan 2024
*St. Francis Xavier University Colloquium	*Baltimore, MD*; Jan 2024
*University of Washington AstroLunch	*Seattle, WA*; Oct 2023
*DESI Collaboration Meeting Plenary	*Cancun, MX*; Dec 2022
*NOIRLab FLASH Talk	*Tucson, AZ*; Nov 2022
*HSC+PFS+Rubin Meeting	*Princeton, NJ*; Nov 2022
*Texas A&M Extragalactic Seminoar	*College Station, TX*; Oct 2022
*University of Texas, Austin Extragalactic Seminar	*Austin, TX*; Oct 2022
*University of Michigan Galaxy Group Seminar	*Ann Arbor, MI*; Sep 2022
Epoch of Galaxy Quenching	*Cambridge, UK*; Sep 2022
A Holistic View of Stellar Feedback and Galaxy Evolution	*Ascona, CH*; July 2022
KIAA Forum on Gas in Galaxies for Early Career Scientists	*Virtual*; Nov 2021
STScI Multi-Object Spectroscopy Workshop	*Virtual*; May 2021
*UMass Amherst Galaxy Lunch	*Amherst, MA*; Apr 2021

Scientific Leadership, Development, and Service

Joint Princeton/IAS Colloquium Series - Organizer	March 2025-Present
Lorentz Center Big Galaxies, Big Problems Workshop - Organizer	Apr 2025

Princeton Galread - Organizer	May 2024-Present
SQuIGG$\vec{L}$E Collaboration - Organizational Lead	Sep 2023-Present
Pitt Galaxy Journal Club - Founding Organizer	Summers 2019, 2020, 2021

Referee for: *Nature, The Astrophysical Journal, Astronomy & Astrophysics, The Open Journal of Astrophysics*
Telescope Allocation Committes: *HST, ALMA (distributed review), NOIRLab*

Science Communication and Teaching

Science Communication:

Guest on WPRB *The Pidgin*: “little red dots”	Dec 2024
Communities Without Walls Continuing Education Guest Speaker	Nov 2024
Sherwood Oaks Retirement Community Continuing Education Guest Speaker	Mar 2023
ACCelerate Festival Presenter: “Making the Largest Maps of the Universe”	Mar 2023
Pittsburgh Public School Research Symposium Judge (2020 Chair of Judging Committee)	Apr 2019, 2020
Steward Observatory 21” telescope operator	Sep 2014-May 2017

Teaching:

Princeton Prison Teaching Initiative Summer Internship, *Lecturer*	Summer 2024
AP Physics C: Mechanics + Electricity & Magnetism, *Tutor*	Acad. Year 19-20
Deitrich School of Arts and Sciences Teaching Assistant Mentor	Acad. Year 18-19
ASTRON 0089: Stars, Galaxies, and Cosmos, *Teaching Assistant*	Spring 2018
Received Myron P. Garfunkel Excellence in Graduate Student Teaching Award	
ASTRON 0088: Stonehenge to Hubble, *Teaching Assistant*	Fall 2017
ASTRON 0087: Basics of Spaceflight, *Teaching Assistant*	Fall 2017
PHYS 141: Introduction to Mechanics, *Preceptor*	Spring 2017
PHYS 241: Introduction to Electricity & Magnetism, *Preceptor*	Spring 2017

Publications

8 first author, **8** second author. As of October 2025, these works have **4290** citations with an h-index of **33**.
* = paper led by student under close supervision of D.S.

First and Second Author:

16. *A Confirmed Deficit of Hot and Cold Dust Emission in the Most Luminous Little Red Dots*
Setton, David J., Greene, Jenny E., Spilker, Justin S., Williams, Christina C., Labbé, Ivo; et al. 2025
The Astrophysical Journal 991, L10

15. **Meet the Neighbors: Gas Rich "Buddy Galaxies" are Common around Recently Quenched Massive Galaxies in the SQuIGG$\vec{L}$E Survey*
Kumar, Anika, **Setton, David J.**, Bezanson, Rachel, Pearl, Alan, Stumbaugh, Erin; et al. 2025
Research Notes of the American Astronomical Society 9, 243

14. *What you see is what you get: empirically measured bolometric luminosities of Little Red Dots*
Greene, Jenny E., **Setton, David J.**, Furtak, Lukas J., Naidu, Rohan P., Volonteri, Marta; et al. 2025
Accepted to the Astrophysical Journal Letters

13. *SQuIGG$\vec{L}$E: Buried star formation cannot explain the rapidly fading CO(2-1) luminosity in massive, $z \sim 0.7$ post-starburst galaxies*
Setton, David J., Spilker, Justin S., Bezanson, Rachel, Suess, Katherine A., Greene, Jenny E.; et al. 2025
Accepted to the Astrophysical Journal

12. **Taking a Break at Cosmic Noon: Continuum-selected Low-mass Galaxies Require Long Burst Cycles*
Mintz, Abby, **Setton, David J.**, Greene, Jenny E., Leja, Joel, Wang, Bingjie; et al. 2025
Submitted to the Astrophysical Journal

11. **UNCOVER: Significant Reddening in Cosmic Noon Quiescent Galaxies*
Siegel, Jared C., **Setton, David J.**, Greene, Jenny E., Suess, Katherine A., Whitaker, Katherine E.; et al. 2025
The Astrophysical Journal 985, 125